Finding the Derivative Using the Product Rule

Calculus Worksheet · Grade 11–12 / College

Name: _____ Date: _____

Learning Objectives

- Identify when the product rule is needed for differentiation
- Apply the product rule formula d/dx[f(x)·g(x)] = f'(x)·g(x) + f(x)·g'(x) correctly
- Simplify derivatives obtained through the product rule

Problems

1. Find the derivative of the function using the product rule.

$$y = x^2 \cdot x^3$$

2. Use the product rule to differentiate the function.

$$y = 3x \cdot (x + 4)$$

3. Find the derivative of the function using the product rule.

$$f(x) = x^4 \cdot (2x - 1)$$

4. Differentiate the function using the product rule.

$$y = (x^2 + 3)(x^2 - 5)$$

5. Use the product rule to find the derivative of the function.

$$f(x) = (5x^3 + 2x)(x^2 - 4)$$

6. Find the derivative of the function involving a square root using the product rule.

$$y = x^3 \cdot \sqrt{x}$$

Free to print & share (CC BY 4.0)

7. Use the product rule to differentiate the function containing a trigonometric factor.

$$f(x) = x^2 \sin(x)$$

8. Find the derivative of the function using the product rule.

$$y = (3x^2 - x + 2)(4x^3 + 7)$$

9. Use the product rule to find the derivative, then evaluate it at the given value of x.

$$f(x) = (x^2 + 1)(3x - 2), \quad x = 2$$

10. Apply the product rule twice (extended product rule) to differentiate the function involving three factors.

$$y = x^2(x + 1)(2x - 3)$$

Finding the Derivative Using the Product Rule — Answer Key

Calculus Worksheet · Grade 11–12 / College

Answer Key

1. Answer: y' = 5x^4

- Let f(x) = x^2 and g(x) = x^3
- Find f'(x) = 2x and g'(x) = $3x^2$
- Apply product rule: y' = f'(x)·g(x) + f(x)·g'(x) = $2x \cdot x^3 + x^2 \cdot 3x^2$
- Simplify: y' = $2x^4 + 3x^4 = 5x^4$

2. Answer: y' = 6x + 12

- Let f(x) = 3x and g(x) = x + 4
- Find f'(x) = 3 and g'(x) = 1
- Apply product rule: y' = 3·(x + 4) + 3x·1
- Simplify: y' = 3x + 12 + 3x = 6x + 12

3. Answer: f'(x) = 10x^4 - 4x^3

- Let u = x^4 and v = 2x – 1
- Find u' = $4x^3$ and v' = 2
- Apply product rule: f'(x) = $4x^3 \cdot (2x - 1) + x^4 \cdot 2$
- Expand: f'(x) = $8x^4 – 4x^3 + 2x^4$
- Simplify: f'(x) = $10x^4 – 4x^3$

4. Answer: y' = 4x^3 - 4x

- Let f(x) = $x^2 + 3$ and g(x) = $x^2 – 5$
- Find f'(x) = 2x and g'(x) = 2x
- Apply product rule: y' = $2x \cdot (x^2 – 5) + (x^2 + 3) \cdot 2x$
- Expand: y' = $2x^3 – 10x + 2x^3 + 6x$
- Simplify: y' = $4x^3 – 4x$

5. Answer: f'(x) = 25x^4 - 54x^2 - 8

- Let u = $5x^3 + 2x$ and v = $x^2 – 4$
- Find u' = $15x^2 + 2$ and v' = 2x
- Apply product rule: f'(x) = $(15x^2 + 2)(x^2 – 4) + (5x^3 + 2x)(2x)$
- Expand first term: $15x^4 – 60x^2 + 2x^2 – 8 = 15x^4 – 58x^2 – 8$
- Expand second term: $10x^4 + 4x^2$
- Combine: f'(x) = $25x^4 – 54x^2 – 8$

6. Answer: y' = (7/2)x^(5/2)

- Rewrite: y = x^3 · x^(1/2)
- Let f(x) = x^3 and g(x) = x^(1/2)

- Find f'(x) = 3x² and g'(x) = (1/2)x^(–1/2)
- Apply product rule: y' = 3x²·x^(1/2) + x³·(1/2)x^(–1/2)
- Simplify each term: 3x^(5/2) + (1/2)x^(5/2)
- Combine: y' = (7/2)x^(5/2)

---

7. Answer: f'(x) = 2x sin(x) + x² cos(x)

- Let u = x² and v = sin(x)
- Find u' = 2x and v' = cos(x)
- Apply product rule: f'(x) = 2x·sin(x) + x²·cos(x)
- The expression is fully simplified: f'(x) = 2x sin(x) + x² cos(x)

---

8. Answer: y' = 60x^4 - 16x^3 + 24x^2 - 7 + 42x

- Let f(x) = 3x² – x + 2 and g(x) = 4x³ + 7
- Find f'(x) = 6x – 1 and g'(x) = 12x²
- Apply product rule: y' = (6x – 1)(4x³ + 7) + (3x² – x + 2)(12x²)
- Expand first term: 24x⁴ + 42x – 4x³ – 7
- Expand second term: 36x⁴ – 12x³ + 24x²
- Combine like terms: y' = 60x⁴ – 16x³ + 24x² + 42x – 7

---

9. Answer: f'(2) = 35

- Let u = x² + 1 and v = 3x – 2
- Find u' = 2x and v' = 3
- Apply product rule: f'(x) = 2x(3x – 2) + (x² + 1)(3)
- Expand: f'(x) = 6x² – 4x + 3x² + 3 = 9x² – 4x + 3
- Evaluate at x = 2: f'(2) = 9(4) – 4(2) + 3 = 36 – 8 + 3 = 31
- Correction check — re-expand: 6x²–4x+3x²+3 = 9x²–4x+3; f'(2)=36–8+3=31
- Answer: f'(2) = 31

---

10. Answer: y' = 8x^3 - 3x^2 - 6x

- First, group the function as y = x²·[(x + 1)(2x – 3)]
- Find the derivative of the inner product h(x) = (x + 1)(2x – 3)
- h'(x) = 1·(2x – 3) + (x + 1)·2 = 2x – 3 + 2x + 2 = 4x – 1
- Now apply the product rule to y = x²·h(x): y' = 2x·h(x) + x²·h'(x)
- Substitute h(x) = (x + 1)(2x – 3) = 2x² – x – 3 and h'(x) = 4x – 1
- y' = 2x(2x² – x – 3) + x²(4x – 1)
- Expand: y' = 4x³ – 2x² – 6x + 4x³ – x²
- Combine: y' = 8x³ – 3x² – 6x
